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A regressor-free adaptive controller for robot manipulators without Slotine and Li's modification

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SUMMARY
Similar to the traditional adaptive strategies for robot manipulators, the regressor-free adaptive controller design also requires applying Slotine and Li's modification to avoid the feedback of joint accelerations. In this paper, a simple method is proposed to construct a regressor-free adaptive controller for robot manipulators without Slotine and Li's modification. In the new design, the joint acceleration vector is lumped into an unknown time-varying function and the function approximation technique is utilized to cover its effect; therefore, its implementation is free from joint acceleration feedback. The closed-loop stability and boundedness of internal signals are justified by the Lyapunov-like technique. Both simulation and experimental results for a two-link robot are presented to show the effectiveness of the proposed design.

KEYWORDS: Robot; Adaptive control; Function approximation technique (FAT).

1. Introduction

Control of robot manipulators is well known to be challenging due to highly nonlinear dynamics in their mathematical models. Traditional (non-adaptive) tracking controllers for robot manipulators, such as the computed-torque controller, are commonly combinations of nonlinear feedforward terms to cover the nonlinear dynamics and linear feedback components to ensure convergence of the error dynamics. However, they might not give proper performance due to various uncertainties and disturbances in the practical robot dynamics. To overcome this problem, some intelligent designs[1–3] have been used to deal with the uncertainties. Neural network-based force control[4,5] that compensates for the unknown dynamics has been proposed. As for adaptive designs,[6–11] the robot dynamics has to be linearly parameterized into a multiplication of a known regressor matrix and an unknown parameter vector. The derivation of the regressor matrix is tedious for a high-DOF (degrees of freedom) robot in general. Since the regressor matrix is configuration dependent, it has to be computed in every control cycle which results in a considerable burden to the control computer. Hence, some regressor-free approaches[12–15] had been suggested. Hsia *et al.*[12] proposed an independent joint controller without using the regressor matrix. Their controller treated the dynamics of robot as a disturbance, and rejected by large gear reductions on the assumption that the gear ratio is sufficient. Song[13] proposed a non-regressor-based approach for an adaptive robot motion controller. This approach lumped system dynamics and uncertainties together, and some bounds need to be known. A robust adaptive controller without using regressor was proposed in Su and Stepanenk,[14] but some bounds of the system dynamics should be available. However, these bounds of the system dynamics are not easy to find for a high-DOF robot. Huang *et al.*[15,16] proposed an adaptive controller based on the function approximation technique (FAT) without using the regressor. The basic idea is to transform the system uncertainties into finite linear combinations of basis functions with unknown constant weightings, therefore, the problem is to find a set of constant weightings. Since the weightings are time invariant, their update law can be easily derived from the conventional Lyapunov design. This approach is available in the compliant motion control applications,[16] the control of the flexible-joint robots.[17] Similar to the traditional adaptive control for robot manipulators, the Slotine and Li modification[11] can be applied to these FAT-based controller designs to avoid the feedback of joint accelerations.

In this paper, we would like to propose a new FAT-based regressor-free adaptive controller for robot manipulators without Slotine and Li's modification so that there is no need for the joint acceleration feedback. This design is an extension of our earlier work[15] where the unknown system dynamics is represented by a finite combination of basis functions weighted with unknown constants. The update laws for these constants can be designed by using the Lyapunov-like technique so that the closed-loop stability and boundedness of internal signals can be ensured. Both simulation and experimental results for a two-link rigid robot are presented to justify the effectiveness of the proposed approach.

In the next section, we review the regressor-based adaptive control of robot manipulators where we see that Slotine and Li's modification can be used to avoid the joint acceleration feedback. Section 3 presents the regressor-free adaptive design with Slotine and Li's modification. Section 4 proposes the new design without Slotine and Li's modification. Simulation and experimental results are shown in Section 5. The last section gives the conclusion.

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2. Review of Regressor-Based Adaptive Control of Robots

Consider an n-link robot manipulator described by the dynamic equation

$$\mathbf{D}(\mathbf{q})\ddot{\mathbf{q}} + \mathbf{C}(\mathbf{q}, \dot{\mathbf{q}})\dot{\mathbf{q}} + \mathbf{g}(\mathbf{q}) = \boldsymbol{\tau}, \tag{1}$$

where $\mathbf{q} \in \Re^n$ is the vector of generalized coordinates, $\mathbf{D}(\mathbf{q}) \in \Re^{n\times n}$ is the inertia matrix, $\mathbf{C}(\mathbf{q}, \dot{\mathbf{q}})\dot{\mathbf{q}} \in \Re^n$ is the vector of centrifugal and Coriolis forces, $\mathbf{g}(\mathbf{q}) \in \Re^n$ is the gravity vector, and $\tau \in \Re^n$ is the control torque vector. If all parameters are available, a feedback linearization controller can be designed as

$$\boldsymbol{\tau} = \mathbf{D}[\ddot{\mathbf{q}}_d - \mathbf{K}_d\dot{\mathbf{e}} - \mathbf{K}_p\mathbf{e}] + \mathbf{C}\dot{\mathbf{q}} + \mathbf{g}, \tag{2}$$

where $\mathbf{q}_d$ is the desired trajectory, $\mathbf{e} = \mathbf{q} - \mathbf{q}_d$ is the tracking error, and matrices $\mathbf{K}_d$, $\mathbf{K}_p \in \Re^{n\times n}$ are selected so that the closed-loop dynamics

$$\ddot{\mathbf{e}} + \mathbf{K}_d\dot{\mathbf{e}} + \mathbf{K}_p\mathbf{e} = \mathbf{0} \tag{3}$$

is asymptotically stable. To implement controller (2), the exact system dynamics is needed. Since a precise robot model is not available, the error dynamics (3) may not converge to zero as desired.

To consider uncertainties in the robot dynamics, an adaptive controller can be designed by assuming that (1) is linearly parameterizable into

$$\mathbf{D}(\mathbf{q})\ddot{\mathbf{q}} + \mathbf{C}(\mathbf{q}, \dot{\mathbf{q}})\dot{\mathbf{q}} + \mathbf{g}(\mathbf{q}) = \mathbf{Y}(\mathbf{q}, \dot{\mathbf{q}}, \ddot{\mathbf{q}})\mathbf{p}, \tag{4}$$

where $\mathbf{Y}(\mathbf{q}, \dot{\mathbf{q}}, \ddot{\mathbf{q}}) \in \Re^{n\times r}$ is a known regressor matrix and $\mathbf{p} \in \Re^r$ is an unknown constant parameter vector. According to Craig *et al.*,[7] an adaptive controller can be designed as

$$\boldsymbol{\tau} = \hat{\mathbf{D}}[\ddot{\mathbf{q}}_d - \mathbf{K}_d\dot{\mathbf{e}} - \mathbf{K}_p\mathbf{e}] + \hat{\mathbf{C}}\dot{\mathbf{q}} + \hat{\mathbf{g}}, \tag{5}$$

where $\hat{\mathbf{D}}$, $\hat{\mathbf{C}}$, and $\hat{\mathbf{g}}$ are estimates of their corresponding quantities. If $\hat{\mathbf{D}}$ is invertible for all $t \geq 0$, then the close loop system becomes

$$\ddot{\mathbf{e}} + \mathbf{K}_d\dot{\mathbf{e}} + \mathbf{K}_p\mathbf{e} = -\hat{\mathbf{D}}^{-1}\mathbf{Y}(\mathbf{q}, \dot{\mathbf{q}}, \ddot{\mathbf{q}})\tilde{\mathbf{p}}, \tag{6}$$

where $\tilde{\mathbf{p}} = \mathbf{p} - \hat{\mathbf{p}}$ and $\hat{\mathbf{p}} \in \Re^r$ is the estimate of $\mathbf{p}$. Suppose a proper update law $\dot{\hat{\mathbf{p}}}$ can be selected so that $\hat{\mathbf{p}} \to \mathbf{p}$ asymptotically, then (6) converges to (3) and we may have tracking error convergence. Let $\mathbf{x} = [\mathbf{e}^T\dot{\mathbf{e}}^T]^T \in \Re^{2n}$ be the state vector, then (6) can be represented in the state space form

$$\dot{\mathbf{x}} = \mathbf{A}\mathbf{x} - \mathbf{B}\hat{\mathbf{D}}^{-1}\mathbf{Y}\tilde{\mathbf{p}}, \tag{7}$$

where $\mathbf{A} = \begin{bmatrix} \mathbf{0} & \mathbf{I}_n \\ -\mathbf{K}_p & -\mathbf{K}_d \end{bmatrix} \in \Re^{2n\times 2n}$, $\mathbf{B} = \begin{bmatrix} \mathbf{0} \\ \mathbf{I}_n \end{bmatrix} \in \Re^{2n\times n}$, and $\mathbf{I}_n \in \Re^{n\times n}$ is an identity matrix. Define a Lyapunov-like function candidate $V(\mathbf{x}, \tilde{\mathbf{p}}) = \frac{1}{2}\mathbf{x}^T\mathbf{P}\mathbf{x} + \frac{1}{2}\tilde{\mathbf{p}}^T\Gamma\tilde{\mathbf{p}}$, where $\boldsymbol{\Gamma} = \boldsymbol{\Gamma}^T \in \Re^{r\times r}$ and $\mathbf{P} = \mathbf{P}^T \in \Re^{2n\times 2n}$ are positive definite matrices. Since $\mathbf{A}$ is Hurwitz, there is a positive definite $\mathbf{Q} = \mathbf{Q}^T \in \Re^{2n\times 2n}$ satisfying $\mathbf{A}^T\mathbf{P} + \mathbf{P}\mathbf{A} = -\mathbf{Q}$. Then, the time derivative of V along the trajectory of (7) is derived as

$$\dot{V} = -\frac{1}{2}\mathbf{x}^T\mathbf{Q}\mathbf{x} - \tilde{\mathbf{p}}^T[(\hat{\mathbf{D}}^{-1}\mathbf{Y})^T\mathbf{B}^T\mathbf{P}\mathbf{x} + \Gamma\dot{\hat{\mathbf{p}}}].$$

Therefore, we may select the update law as

$$\dot{\hat{\mathbf{p}}} = -\Gamma^{-1}[\hat{\mathbf{D}}^{-1}\mathbf{Y}(\mathbf{q}, \dot{\mathbf{q}}, \ddot{\mathbf{q}})]^T\mathbf{B}^T\mathbf{P}\mathbf{x}. \tag{8}$$

Hence, we have $\dot{V} = -\frac{1}{2}\mathbf{x}^T\mathbf{Q}\mathbf{x} \leq 0$. To implement the update law, we need to ensure $\hat{\mathbf{D}}$ to be invertible for all $t \geq 0$, and we need to feed back $\ddot{\mathbf{q}}$, i.e., joint acceleration, in real time. To overcome these problems, Slotine and Li proposed the following approach.[11] Define an error vector $\mathbf{s} = \dot{\mathbf{e}} + \Lambda\mathbf{e}$ where $\Lambda = \mathrm{diag}(\lambda_1, \lambda_2, ..., \lambda_n) \in \Re^{n\times n}$ with $\lambda_i > 0 \in \Re$ for all $i = 1, \ldots, n$. Hence, the robot model (1) can be rewritten as

$$\mathbf{D}\dot{\mathbf{s}} + \mathbf{C}\mathbf{s} + \mathbf{g} + \mathbf{D}\ddot{\mathbf{q}}_d - \mathbf{D}\Lambda\dot{\mathbf{e}} + \mathbf{C}\dot{\mathbf{q}}_d - \mathbf{C}\ \Lambda\mathbf{e} = \tau. \tag{9}$$

Define the signal vector $\mathbf{v} = \dot{\mathbf{q}}_d - \Lambda\mathbf{e}$ and we may design a controller as

$$\boldsymbol{\tau} = \hat{\mathbf{D}}\dot{\mathbf{v}} + \hat{\mathbf{C}}\mathbf{v} + \hat{\mathbf{g}} - \mathbf{K}_d\mathbf{s}, \tag{10}$$

where $\mathbf{K}_d \in \Re^{n\times n}$ is a positive definite matrix. Then, the closed-loop dynamics becomes

$$\mathbf{D}\dot{\mathbf{s}} + \mathbf{C}\mathbf{s} + \mathbf{K}_d\mathbf{s} = -\tilde{\mathbf{D}}\dot{\mathbf{v}} - \tilde{\mathbf{C}}\mathbf{v} - \tilde{\mathbf{g}}, \tag{11}$$

where $\tilde{\mathbf{D}} = \mathbf{D} - \hat{\mathbf{D}}$, $\tilde{\mathbf{C}} = \mathbf{C} - \hat{\mathbf{C}}$, and $\tilde{\mathbf{g}} = \mathbf{g} - \hat{\mathbf{g}}$ are estimation errors. We may thus represent (11) into the linearly parameterized form

$$\mathbf{D}\dot{\mathbf{s}} + \mathbf{C}\mathbf{s} + \mathbf{K}_d\mathbf{s} = -\mathbf{Y}_1(\mathbf{q}, \dot{\mathbf{q}}, \mathbf{v}, \dot{\mathbf{v}})\tilde{\mathbf{p}}_1, \tag{12}$$

where $\mathbf{Y}_1$ is the regressor matrix and $\tilde{\mathbf{p}}_1 = \mathbf{p}_1 - \hat{\mathbf{p}}_1$. Selecting the Lyapunov-like function candidate $V(\mathbf{s}, \tilde{\mathbf{p}}_1) = \frac{1}{2}\mathbf{s}^T\mathbf{D}\mathbf{s} + \frac{1}{2}\tilde{\mathbf{p}}_1^T\Gamma\tilde{\mathbf{p}}_1$, and using the fact that the matrix $\dot{\mathbf{D}} - 2\mathbf{C}$ is skew symmetric, we may have $\dot{V} = -\mathbf{s}^T\mathbf{K}_d\mathbf{s} - \tilde{\mathbf{p}}_1^T(\Gamma\dot{\hat{\mathbf{p}}}_1 + \mathbf{Y}_1^T\mathbf{s})$. Therefore, if the update law is chosen to be

$$\dot{\hat{\mathbf{p}}}_1 = -\boldsymbol{\Gamma}^{-1}\mathbf{Y}_1^T(\mathbf{q}, \dot{\mathbf{q}}, \mathbf{v}, \dot{\mathbf{v}})\mathbf{s}, \tag{13}$$

then we have $\dot{V} = -\mathbf{s}^T\mathbf{K}_d\mathbf{s} \leq 0$. It is seen that to implement the update law, we do not have the singularity problem for $\hat{\mathbf{D}}$, nor do we need to feed back the joint acceleration. However, to implement (13), we still need to go through a tedious derivation for the regressor matrix and compute its value in every control cycle.

3. Review of Regressor-Free Adaptive Control of Robots

The regressor-free design starts from the robot model (9). The FAT[15–17] is employed to represent $\mathbf{D}$, $\mathbf{C}$, and $\mathbf{g}$ by

$$\mathbf{D} = \mathbf{W}_\mathbf{D}^T\mathbf{Z}_\mathbf{D}, \mathbf{C} = \mathbf{W}_\mathbf{C}^T\mathbf{Z}_\mathbf{C}, \mathbf{g} = \mathbf{W}_\mathbf{g}^T\mathbf{z}_\mathbf{g},$$

where $\mathbf{W}_\mathbf{D} \in \Re^{n^2\beta_D\times n}$, $\mathbf{W}_\mathbf{C} \in \Re^{n^2\beta_C\times n}$, and $\mathbf{W}_\mathbf{g} \in \Re^{n\beta_g\times n}$ are weighting matrices and $\mathbf{Z}_\mathbf{D} \in \Re^{n^2\beta_D\times n}$, $\mathbf{Z}_\mathbf{C} \in \Re^{n^2\beta_C\times n}$, and

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$\mathbf{z_g} \in \Re^{n\beta_g \times 1}$ are matrices of basis functions. The number $\beta_{(\cdot)}$ represents the number of basis functions used, which is selected large enough so that the approximation error can be ignored. Using the same set of basis functions, the corresponding estimates can also be represented as

$$\hat{\mathbf{D}} = \hat{\mathbf{W}}_{\mathbf{D}}^T \mathbf{Z_D}, \hat{\mathbf{C}} = \hat{\mathbf{W}}_{\mathbf{C}}^T \mathbf{Z_C}, \hat{\mathbf{g}} = \hat{\mathbf{W}}_{\mathbf{g}}^T \mathbf{z_g}.$$

Hence, controller (10) is rewritten as

$$\boldsymbol{\tau} = \hat{\mathbf{W}}_{\mathbf{D}}^T \mathbf{Z_D} \dot{\mathbf{v}} + \hat{\mathbf{W}}_{\mathbf{C}}^T \mathbf{Z_C} \mathbf{v} + \hat{\mathbf{W}}_{\mathbf{g}}^T \mathbf{z_g} - \mathbf{K}_d \mathbf{s} \tag{14}$$

and the error dynamics becomes

$$\mathbf{D}\dot{\mathbf{s}} + \mathbf{C}\mathbf{s} + \mathbf{K}_d \mathbf{s} = -\tilde{\mathbf{W}}_{\mathbf{D}}^T \mathbf{Z_D} \dot{\mathbf{v}} - \tilde{\mathbf{W}}_{\mathbf{C}}^T \mathbf{Z_C} \mathbf{v} - \tilde{\mathbf{W}}_{\mathbf{g}}^T \mathbf{z_g},$$

where $\tilde{\mathbf{W}}_{(\cdot)} = \mathbf{W}_{(\cdot)} - \hat{\mathbf{W}}_{(\cdot)}$. Define a Lyapunov-like function candidate

$$V = \frac{1}{2}\mathbf{s}^T \mathbf{D}\mathbf{s} + \frac{1}{2} Tr\left(\tilde{\mathbf{W}}_{\mathbf{D}}^T \mathbf{Q_D} \tilde{\mathbf{W}}_{\mathbf{D}} + \tilde{\mathbf{W}}_{\mathbf{C}}^T \mathbf{Q_C} \tilde{\mathbf{W}}_{\mathbf{C}} + \tilde{\mathbf{W}}_{\mathbf{g}}^T \mathbf{Q_g} \tilde{\mathbf{W}}_{\mathbf{g}}\right),$$

where $\mathbf{Q}_{(\cdot)}$ are symmetric positive definite matrices. The update laws can be selected as

$$\begin{aligned} \dot{\hat{\mathbf{W}}}_{\mathbf{D}} &= -\mathbf{Q}_{\mathbf{D}}^{-1} \mathbf{Z_D} \dot{\mathbf{v}} \mathbf{s}^T \quad \dot{\hat{\mathbf{W}}}_{\mathbf{C}} = -\mathbf{Q}_{\mathbf{C}}^{-1} \mathbf{Z_C} \mathbf{v} \mathbf{s}^T \\ \dot{\hat{\mathbf{W}}}_{\mathbf{g}} &= -\mathbf{Q}_{\mathbf{g}}^{-1} \mathbf{z_g} \mathbf{s}^T \end{aligned} \tag{15}$$

so that the time derivative of V is written as $\dot{V} = -\mathbf{s}^T \mathbf{K}_d \mathbf{s} \leq 0$. The convergence of the output error can further be proved by Barbalat's lemma. It is seen that the joint acceleration is not needed in (15) which is due to the application of the Slotine and Li modification.

Remark. In the above reviews, we firstly derive a conventional adaptive controller for robot manipulators. To realize the update law, the estimate of the inertia matrix is required to be nonsingular and the joint accelerations should be known. Afterward, the Slotine and Li approach is derived to eliminate the requirement for the acceleration feedback and avoid the singularity problem. However, both approaches need to calculate the regressor matrix. In Section 3, a regressor-free adaptive controller is derived based on FAT. Nevertheless, it is still based on the Slotine and Li modification. In the next section, we are going to propose a new regressor-free adaptive controller without Slotine and Li's modification.

4. Main Results

Let us consider system model (1) again. Since **D**, **C**, and **g** are functions of states and hence functions of time, the robot dynamics can be regarded as an unknown time-varying function. In this design, we would like to lump the joint acceleration vector and the robot dynamics together. Therefore, we may define the unknown time-varying function $\boldsymbol{\Psi}(t) = \mathbf{D}\ddot{\mathbf{q}} + \mathbf{C}\dot{\mathbf{q}} + \mathbf{g} - \ddot{\mathbf{q}}$, and a controller is proposed as

$$\boldsymbol{\tau} = \hat{\boldsymbol{\Psi}} + \ddot{\mathbf{q}}_d - \mathbf{K}_d \dot{\mathbf{e}} - \mathbf{K}_p \mathbf{e}, \tag{16}$$

where $\hat{\boldsymbol{\Psi}} \in \Re^n$ is the estimate of $\boldsymbol{\Psi}$. Then, the closed-loop dynamics becomes

$$\ddot{\mathbf{e}} + \mathbf{K}_d \dot{\mathbf{e}} + \mathbf{K}_p \mathbf{e} = \hat{\boldsymbol{\Psi}} - \boldsymbol{\Psi}. \tag{17}$$

If a proper update law for $\hat{\boldsymbol{\Psi}}$ can be found to have $\hat{\boldsymbol{\Psi}} \to \boldsymbol{\Psi}$, then the tracking error **e** converges asymptotically. To design the update law, let us apply the FAT to represent $\boldsymbol{\Psi}$ and $\hat{\boldsymbol{\Psi}}$ as

$$\boldsymbol{\Psi} = \mathbf{W}^T \mathbf{z} + \boldsymbol{\varepsilon}, \tag{18a}$$

$$\hat{\boldsymbol{\Psi}} = \hat{\mathbf{W}}^T \mathbf{z}, \tag{18b}$$

where $\mathbf{W} \in \Re^{n\beta \times n}$ is the weighting matrix, $\mathbf{z}(t) \in \Re^{n\beta \times 1}$ is the vector of basis functions, and ε is the approximation error. With these representations, we may rewrite (17) to be

$$\ddot{\mathbf{e}} + \mathbf{K}_d \dot{\mathbf{e}} + \mathbf{K}_p \mathbf{e} = -\tilde{\mathbf{W}}^T \mathbf{z} - \boldsymbol{\varepsilon} \tag{19}$$

or, in the state space form

$$\dot{\mathbf{x}} = \mathbf{A}\mathbf{x} - \mathbf{B}(\tilde{\mathbf{W}}^T \mathbf{z} + \boldsymbol{\varepsilon}), \tag{20}$$

where **x**, **A**, and **B** are the same as those in (7). Now, let us consider the Lyapunov-like function candidate

$$V = \frac{1}{2}\mathbf{x}^T \mathbf{P}\mathbf{x} + \frac{1}{2} Tr(\tilde{\mathbf{W}}^T \boldsymbol{\Gamma} \tilde{W}), \tag{21}$$

where $\boldsymbol{\Gamma} = \boldsymbol{\Gamma}^T \in \Re^{n\beta \times n\beta}$ is a positive definite matrix and $\mathbf{P} = \mathbf{P}^T \in \Re^{2n \times 2n}$ is a positive definite matrix satisfying the Lyapunov equation $\mathbf{A}^T \mathbf{P} + \mathbf{P}\mathbf{A} = -\mathbf{Q}$ for a given positive definite and symmetric matrix $\mathbf{Q} \in \Re^{2n \times 2n}$. Taking the time derivative of V along the trajectory of (20), and using the cyclic property of the trace operation,[16] we have

$$\dot{V} = -\frac{1}{2}\mathbf{x}^T \mathbf{Q}\mathbf{x} - \mathbf{x}^T \mathbf{P}\mathbf{B}\boldsymbol{\varepsilon} - Tr(\tilde{\mathbf{W}}^T \mathbf{z}\mathbf{x}^T \mathbf{P}\mathbf{B} + \tilde{\mathbf{W}}^T \Gamma \dot{\hat{\mathbf{W}}}). \tag{22}$$

Therefore, the update law with σ-modification can be designed as

$$\dot{\hat{\mathbf{W}}} = -\boldsymbol{\Gamma}^{-1}(\mathbf{z}\mathbf{x}^T \mathbf{P}\mathbf{B} + \sigma \hat{\mathbf{W}}), \tag{23}$$

where σ is a positive number. If a sufficient number of basis functions are used so that the approximation error ε can be ignored and if there is no σ-modification, then we may have $\dot{V} = -\frac{1}{2}\mathbf{x}^T \mathbf{Q}\mathbf{x} \leq 0$. Hence, the convergence of **x** can be further proved by Barbalat's lemma. On the other hand, with the presence of ε and σ, we may still prove that **x** is uniformly ultimately bounded (Appendix A) and the transient response is bounded by a weighted exponential function shifted by some constant (Appendix B). It is seen that the realization of (23) is free from the computation of the regressor matrix. To avoid the feedback of joint accelerations, this approach does not need the support of Slotine and Li's modification. The adaptive control laws derived in this paper are summarized in Table I to facilitate the understanding of the improvements in the controller design.

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Table I. Summary of the adaptive controllers for robots.

	Conventional	Slotine & Li's	Regressor-free	New design
Controller	(5)	(10)	(14)	(16)
Update law	$\dot{\hat{\mathbf{p}}} = -\mathbf{\Gamma}^{-1}[\hat{\mathbf{D}}^{-1}\mathbf{Y}(\mathbf{q}, \dot{\mathbf{q}}, \ddot{\mathbf{q}})]^T\mathbf{B}^T\mathbf{P}\mathbf{x}$	$\dot{\hat{\mathbf{p}}} = -\mathbf{\Gamma}^{-1}\mathbf{Y}_1^T(\mathbf{q}, \dot{\mathbf{q}}, \mathbf{v}, \dot{\mathbf{v}})\mathbf{s}$	$\dot{\hat{\mathbf{W}}}_\mathbf{D} = -\mathbf{Q}_\mathbf{D}^{-1}\mathbf{Z}_\mathbf{D}\dot{\mathbf{v}}\mathbf{s}^T$ $\dot{\hat{\mathbf{W}}}_\mathbf{C} = -\mathbf{Q}_\mathbf{C}^{-1}\mathbf{Z}_\mathbf{C}\mathbf{v}\mathbf{s}^T$ $\dot{\hat{\mathbf{W}}}_\mathbf{g} = -\mathbf{Q}_\mathbf{g}^{-1}\mathbf{z}_\mathbf{g}\mathbf{s}^T$	$\dot{\hat{\mathbf{W}}} = -\mathbf{\Gamma}^{-1}(\mathbf{z}\mathbf{x}^T\mathbf{P}\mathbf{B} + \sigma\hat{\mathbf{W}})$
Realization issue				
$\hat{\mathbf{D}}^{-1}$	Needed	Not needed	Not needed	Not needed
$\ddot{\mathbf{q}}$	Needed	Not needed	Not needed	Not needed
Regressor	Needed	Needed	Not needed	Not needed
Slotine–Li Modification	Not needed	Needed	Needed	Not needed

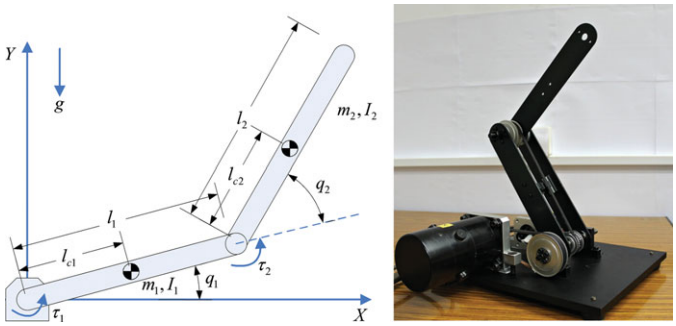

Fig. 1. (Colour online) Experimental setup.

5. Simulation and Experimental Results

In this section, computer simulations and experiments are performed to justify the effectiveness of the proposed approach. The experimental system is introduced in Section 5.1. Simulation results are presented in Section 5.2. Experimental results are given in Section 5.3.

5.1. Experiment setup

A 2D robot is constructed as shown in Fig. 1. Computer simulations and experiments are performed with the following robot parameters:$m_1 = 0.55\,\text{kg}$, $m_2 = 0.23\,\text{kg}$, $l_1 = l_2 = 0.2\,\text{m}$, $l_{c1} = 0.1\,\text{m}$, $l_{c2} = 0.05\,\text{m}$, $I_1 = 0.00138\ \text{kg}\,\text{m}^2$, and $I_2 = 0.00744\,\text{kg}\,\text{m}^2$. The joints 1 and 2 are driven by 200 W and 80 W DC servo motors, respectively. Two 1000 PPR encoders are installed to feed back the joint angles. The HCTL-2032 decoder provides 4 × count mode to sample the encoder reading. An ADLINK's ACL-8112 AD/DA-card is used to interface with a PC-compatible. The proposed controller is implemented in the timer interrupt service routine at 0.5 ms sampling period under MS-DOS environment.

5.2. Simulation results

The initial condition for the robot is arranged as $\mathbf{q}(0) = \dot{\mathbf{q}}(0) = \mathbf{0}$ so that the end-point is initially at (0.4 m, 0 m) in the Cartesian space. To verify the performance of the controller (16), we would like the end-point to track a 0.06 m radius circle centered at (0.1 m, 0.3 m) in 2 s. Since the initial end-point position is away from the initial desired position (0.1 m, 0.24 m), a significant transient can be observed. The desired trajectory in a circle is to verify the tracking performance of the robot with sufficient difficulties. The controller parameters are selected as

$$\mathbf{K}_p = \begin{bmatrix} 600 & 0 \\ 0 & 200 \end{bmatrix}, \quad \mathbf{K}_d = \begin{bmatrix} 50 & 0 \\ 0 & 50 \end{bmatrix}.$$

The first 31 terms of the Fourier series are employed as the basis functions,[16] i.e., $\mathbf{z}_{(\cdot)} = [1 \sin\omega_1 t \cos\omega_1 t \cdots \sin\omega_n t \cos\omega_n t]^\text{T}$, $\omega_n = 2n\pi/T$, $n \in N$. Therefore, $\mathbf{z} = [\mathbf{z}_1^T\ \mathbf{z}_2^T]^T \in \Re^{62\times 1}$, $\hat{\mathbf{W}} = \text{diag}\{\hat{\mathbf{w}}_1, \hat{\mathbf{w}}_2\} \in \Re^{62\times 2}$. Initial weighting vectors are chosen as

$$\hat{\mathbf{w}}_1(0) = \hat{\mathbf{w}}_2(0) = \mathbf{0} \in \Re^{31\times 1}.$$

The adaptive gain matrices are selected as

$$\mathbf{\Gamma}^{-1} = \mathbf{I}_{62}, \mathbf{P} = \begin{bmatrix} \mathbf{p}_{11} & \mathbf{p}_{12} \\ \mathbf{p}_{21} & \mathbf{p}_{22} \end{bmatrix},$$

$\mathbf{p}_{11} = \mathbf{p}_{22} = \left[\begin{smallmatrix} 100 & 0 \\ 0 & 200 \end{smallmatrix}\right]$, $\mathbf{p}_{12} = \mathbf{p}_{21} = \left[\begin{smallmatrix} 2000 & 0 \\ 0 & 4000 \end{smallmatrix}\right]$, and the σ-modification parameter is $\sigma = 1$.

The simulation results are shown in Figs. 2–6. Figure 2 shows the tracking performance in the Cartesian space. It can be observed that the end-point trajectory converges smoothly to the desired trajectory, although we do not know precise dynamics of the robot. Figure 3 is the joint space tracking performance. The transient converges very quickly. In Figs. 4 and 5, we can see two phases of the estimates: transient and steady state. Since there is a large initial error, a significant transient can be seen in Fig. 4, however, the estimates are bounded as desired and converge very quickly. The steady-state phase is magnified in Fig. 5. It can be seen that $\hat{\Psi}$ converges to Ψ. Figure 6 is the control effort.

5.3. Experimental results

The same set of controller parameters are used in experiments. The experimental results are shown in Figs. 7–11. Figure 7 and 8 show the tracking performance in the Cartesian space and joint space, respectively. The transient converges within 0.4 s and the steady-state error is about 0.01 rad. Figure 9 is the time histories of the estimates. Figure 10 is the control effort. To verify the performance of higher frequency of movement, we would like the end-point to track

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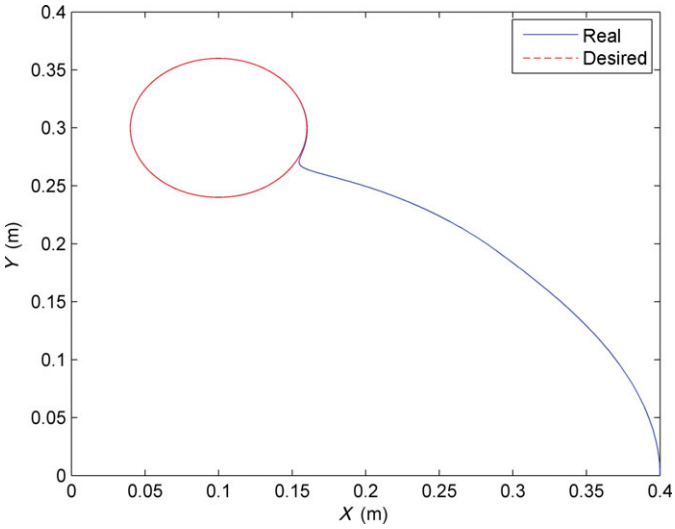

Fig. 2. (Colour online) The Cartesian space tracking performance in simulation.

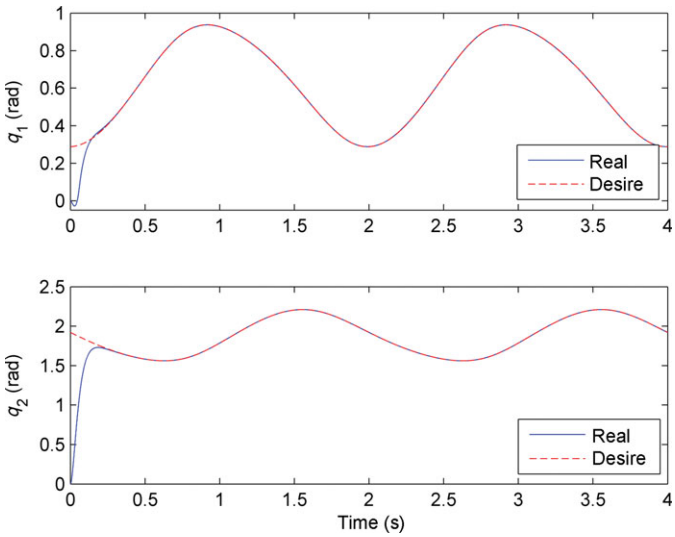

Fig. 3. (Colour online) The joint space tracking performance in simulation.

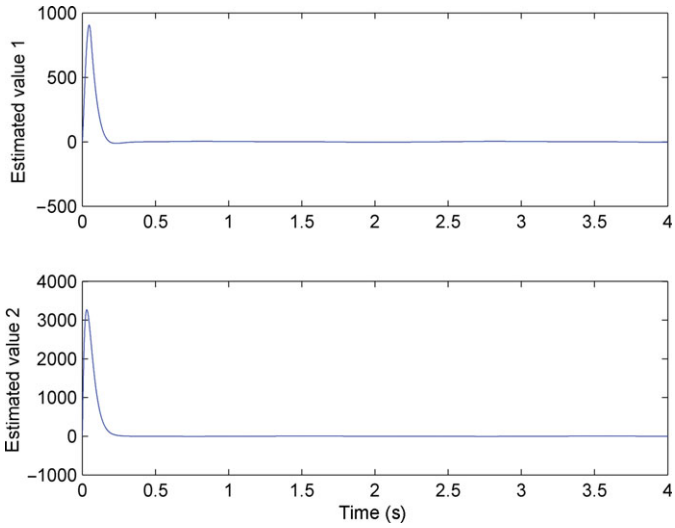

Fig. 4. (Colour online) Approximation of $\mathbf{\Psi}$ in simulation: transient response.

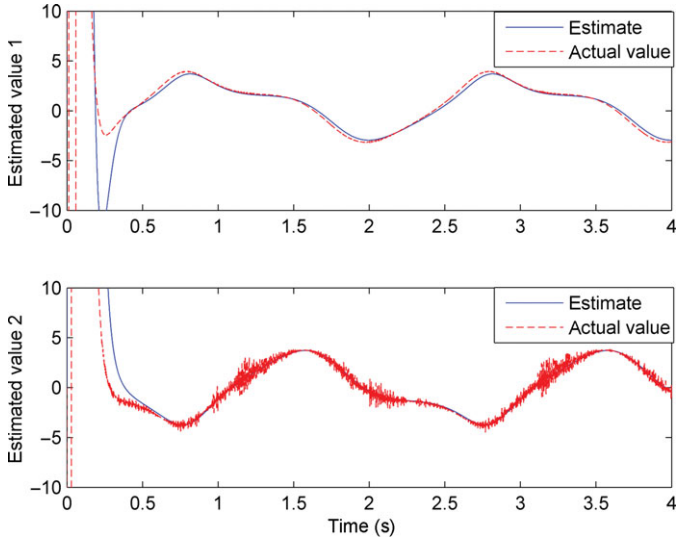

Fig. 5. (Colour online) Approximation of $\mathbf{\Psi}$ in simulation: steady-state response.

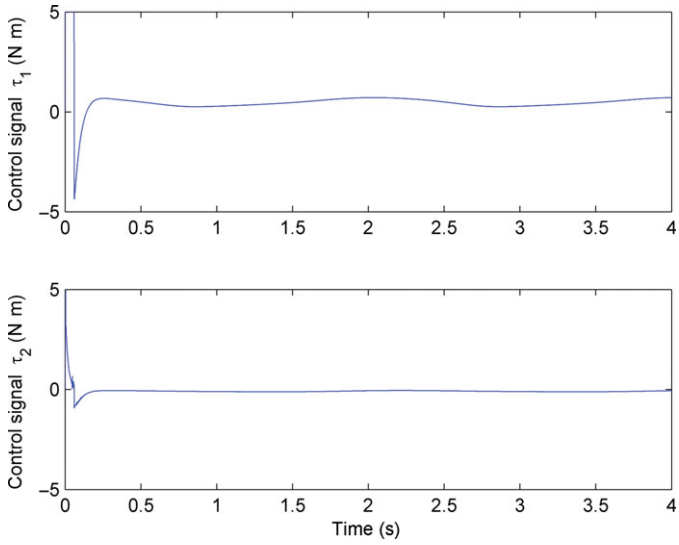

Fig. 6. (Colour online) Control effort in simulation.

the same circle in 1 s. In Fig. 11, it can be observed that the tracking performance in the joint space is still satisfactory.

It is worth noting that to implement the proposed controller, we do not need to know the precise dynamics of the robot. There is no need for the computation of the regressor either. The feedback of the joint accelerations is avoided without using Slotine and Li's modification. All we have to do is to select some controller parameters and some initial weighting vectors. On the other hand, compared with our earlier work in ref. [15], described by the control law (14), the proposed controller (16) is simpler, and its complexity only grows linearly with the increase of the degrees of freedom (Table II). Therefore, a smaller sampling time can be achieved to give better performance, and a lower cost controller can be applied to a high-DOF robot. Since the new design can easily be used to cover the system dynamics, it is obviously applicable to other dynamic systems.

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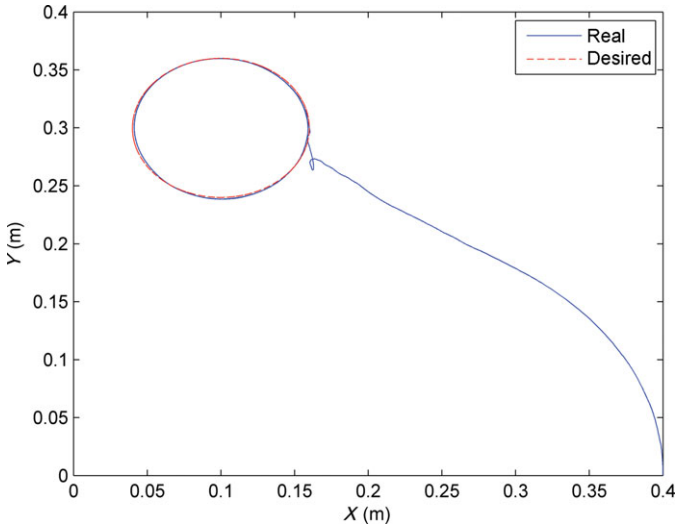

Fig. 7. (Colour online) The Cartesian space tracking performance in the experiment.

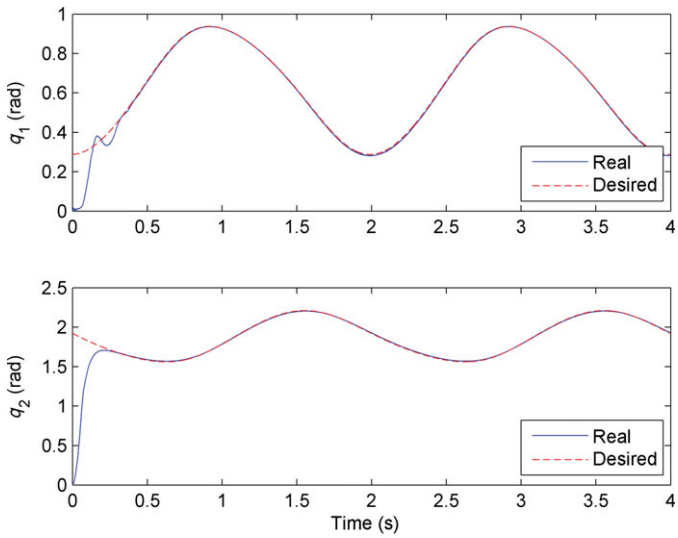

Fig. 8. (Colour online) The joint space tracking performance in the experiment.

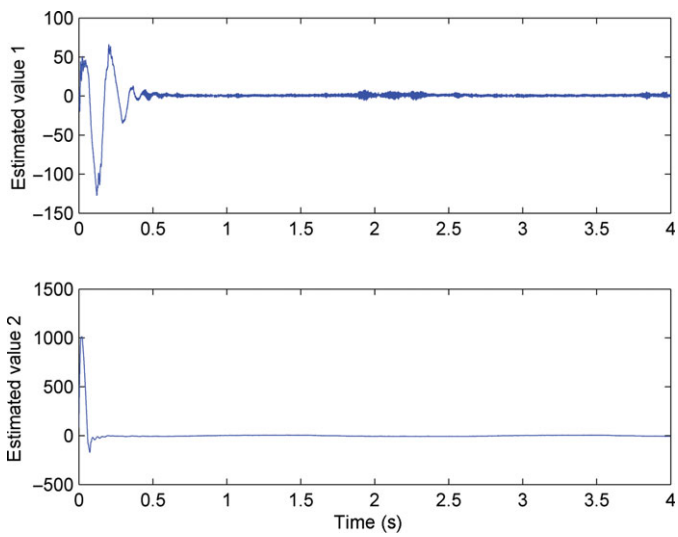

Fig. 9. (Colour online) Approximation of $\mathbf{\Psi}$ in the experiment.

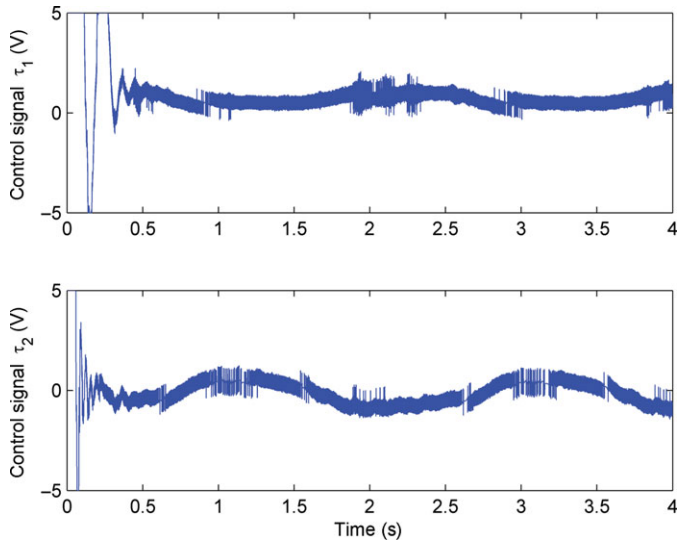

Fig. 10. (Colour online) Control effort in the experiment.

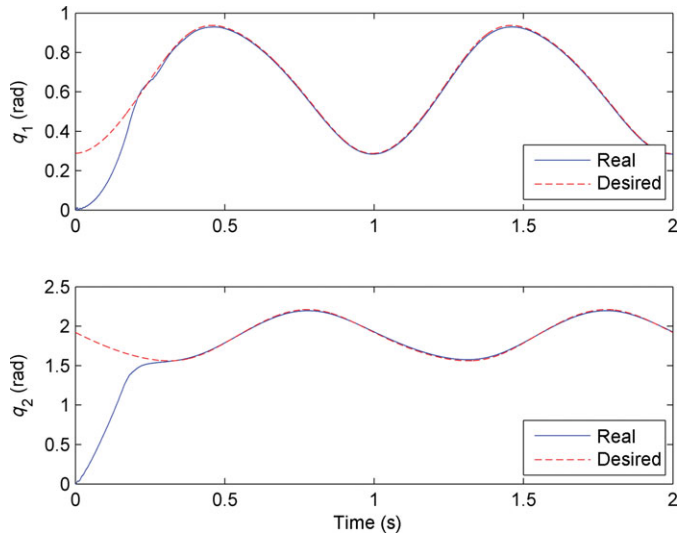

Fig. 11. (Colour online) The joint space tracking performance with higher frequency movement.

Table II. Complexity of the regressor-free adaptive controller.

	With Slotine–Li modification	New design
Controller	(14)	(16)
Estimates in matrix form	$\hat{\mathbf{D}} \in \Re^{n\times n}$	
	$\hat{\mathbf{C}} \in \Re^{n\times n}$	$\hat{\mathbf{\Psi}} \in \Re^{n}$
	$\hat{\mathbf{g}} \in \Re^{n}$	
Number of estimates in terms of matrix entries		
2-DOF robot	10	2
4-DOF robot	36	4
6-DOF robot	78	6
n-DOF robot	$2n^2 + n$	n

6. Conclusions

We have proposed a regressor-free adaptive controller for robot manipulators without using Slotine and Li's modification. Also, for the implementation of the control

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strategy, there is no need of the joint acceleration feedback. The closed-loop stability and boundedness of internal signals have been proved by using the Lyapunov-like technique. Both simulation and experimental results show that the proposed design can give good tracking performance regardless of various uncertainties in the system dynamics.

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Appendix A: Consideration of the approximation error

Here, we would like to consider the effect of the approximation error on the stability of theclosed-loop system. With the update law in (23), (22) becomes

$$\begin{aligned}\dot{V} &= -\frac{1}{2}\mathbf{x}^T\mathbf{Q}\mathbf{x} - \mathbf{x}^T\mathbf{P}\mathbf{B}\boldsymbol{\varepsilon} + \sigma Tr(\tilde{\mathbf{W}}^T\hat{\mathbf{W}}) \\ &\le -\frac{1}{2}\lambda_{\min}(\mathbf{Q})\,\|\mathbf{x}\|^2 + \lambda_{\max}(\mathbf{P})\,\|\boldsymbol{\varepsilon}\|\,\|\mathbf{x}\| + \sigma Tr(\tilde{\mathbf{W}}^T\hat{\mathbf{W}}).\end{aligned}$$

By using the following inequality[16]

$$\begin{aligned}&-\frac{1}{2}\lambda_{\min}(\mathbf{Q})\,\|\mathbf{x}\|^2 + \lambda_{\max}(\mathbf{P})\,\|\boldsymbol{\varepsilon}\|\,\|\mathbf{x}\| \\ &= -\frac{1}{2}\left(\sqrt{\tfrac{1}{2}\lambda_{\min}(\mathbf{Q})}\,\|\mathbf{x}\| - \frac{\lambda_{\max}(\mathbf{P})\,\|\boldsymbol{\varepsilon}\|}{\sqrt{\tfrac{1}{2}\lambda_{\min}(\mathbf{Q})}}\right)^2 \\ &\quad -\frac{1}{2}\left(\frac{1}{2}\lambda_{\min}(\mathbf{Q})\,\|\mathbf{x}\|^2 - \frac{\lambda_{\max}^2(\mathbf{P})}{\frac{1}{2}\lambda_{\min}(\mathbf{Q})}\,\|\boldsymbol{\varepsilon}\|^2\right) \\ &\le -\frac{1}{4}\lambda_{\min}(\mathbf{Q})\,\|\mathbf{x}\|^2 + \frac{\lambda_{\max}^2(\mathbf{P})}{\lambda_{\min}(\mathbf{Q})}\,\|\boldsymbol{\varepsilon}\|^2\end{aligned}$$

$$\begin{aligned}Tr(\tilde{\mathbf{W}}^T\hat{\mathbf{W}}) &= Tr(\tilde{\mathbf{W}}^T\mathbf{W}) - Tr(\tilde{\mathbf{W}}^T\tilde{\mathbf{W}}) \\ &\le \sum_{i=1}^{n}(\|\tilde{\mathbf{w}}_i\|\,\|\mathbf{w}_i\| - \|\tilde{\mathbf{w}}_i\|^2) \\ &= \frac{1}{2}\sum_{i=1}^{n}[\|\mathbf{w}_i\|^2 - \|\tilde{\mathbf{w}}_i\|^2 - (\|\tilde{\mathbf{w}}_i\| - \|\mathbf{w}_i\|)^2] \\ &\le \frac{1}{2}\sum_{i=1}^{n}(\|\mathbf{w}_i\|^2 - \|\tilde{\mathbf{w}}_i\|^2) \\ &= \frac{1}{2}Tr(\mathbf{W}^T\mathbf{W}) - \frac{1}{2}Tr(\tilde{\mathbf{W}}^T\tilde{\mathbf{W}})\end{aligned}$$

and the relationship

$$V \le \frac{1}{2}\lambda_{\max}(\mathbf{P})\,\|\mathbf{x}\|^2 + \frac{1}{2}\lambda_{\max}(\Gamma)Tr(\tilde{\mathbf{W}}^T\tilde{\mathbf{W}}),$$

we may further have

$$\begin{aligned}\dot{V} &\le -\alpha V + \frac{1}{2}[\alpha\lambda_{\max}(\mathbf{P}) - \frac{1}{2}\lambda_{\min}(\mathbf{Q})]\,\|\mathbf{x}\|^2 \\ &\quad + \frac{1}{2}[\alpha\lambda_{\max}(\Gamma) - \sigma]Tr(\tilde{\mathbf{W}}^T\tilde{\mathbf{W}}) + \frac{1}{2}\sigma Tr(\mathbf{W}^T\mathbf{W}) \\ &\quad + \frac{\lambda_{\max}^2(\mathbf{P})}{\lambda_{\min}(\mathbf{Q})}\,\|\boldsymbol{\varepsilon}\|^2.\end{aligned}$$

Selecting $\alpha \le \min\{\frac{\lambda_{\min}(\mathbf{Q})}{2\lambda_{\max}(\mathbf{P})}, \frac{\sigma}{\lambda_{\max}(\Gamma)}\}$, the above inequality can be rewritten as

$$\dot{V} \le -\alpha V + \frac{1}{2}\sigma Tr(\mathbf{W}^T\mathbf{W}) + \frac{\lambda_{\max}^2(\mathbf{P})}{\lambda_{\min}(\mathbf{Q})}\,\|\boldsymbol{\varepsilon}\|^2.$$

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Therefore, $\dot{V} < 0$ whenever

$$(\mathbf{x}, \tilde{\mathbf{W}}) \in \left\{ (\mathbf{x}, \tilde{\mathbf{W}}) \middle| V > \frac{1}{2\alpha}\sigma Tr(\mathbf{W}^T\mathbf{W}) + \frac{\lambda_{\max}^2(\mathbf{P})}{\alpha\lambda_{\min}(\mathbf{Q})} \sup_{\tau \geq t_0} \|\boldsymbol{\varepsilon}(\tau)\|^2 \right\}.$$

This implies that $\mathbf{x}$ and $\tilde{\mathbf{W}}$ are uniformly ultimately bounded.

Appendix B: Transient performance analysis

The upper bound for V can be found as

$$V(t) \leq e^{-\alpha(t-t_0)}V(t_0) + \frac{1}{2\alpha}\sigma Tr(\mathbf{W}^T\mathbf{W}) + \frac{\lambda_{\max}^2(\mathbf{P})}{\alpha\lambda_{\min}(\mathbf{Q})} \sup_{t_0<\tau<t} \|\boldsymbol{\varepsilon}(\tau)\|^2 .$$

According to the relationship

$$V \geq \frac{1}{2}\lambda_{\min}(\mathbf{P})\,\|\mathbf{x}\|^2 + \frac{1}{2}\lambda_{\min}(\Gamma)Tr(\tilde{\mathbf{W}}^T\tilde{\mathbf{W}}),$$

the upper bound for $\|\mathbf{x}\|^2$ can be estimated to be

$$\|\mathbf{x}\|^2 \leq \frac{2V}{\lambda_{\min}(\mathbf{P})}.$$

Hence, we have

$$\|\mathbf{x}\| \leq \sqrt{\frac{2V(t_0)}{\lambda_{\min}(\mathbf{P})}}e^{-\frac{\alpha}{2}(t-t_0)} + \sqrt{\frac{\sigma Tr(\mathbf{W}^T\mathbf{W})}{\alpha\lambda_{\min}(\mathbf{P})}} + \sqrt{\frac{2}{\alpha\lambda_{\min}(\mathbf{Q})\lambda_{\min}(\mathbf{P})}}\lambda_{\max}(\mathbf{P}) \sup_{t_0<\tau<t} \|\boldsymbol{\varepsilon}(\tau)\| .$$

This implies that the transient response is bounded by a weighted exponential function shifted by some constant.

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